SAR Movement Dossier Protocol & Standard Framework

A Standardized Four-Layer Trajectory Analysis Framework for Missing Person Investigations

This framework establishes the mandatory four-component protocol required whenever a **Movement Dossier & Trajectory Analysis** is requested for a missing person case in wilderness, alpine, or resort environments.

Layer / Module	Primary Objective	Key Operational Outputs
1. Multi-Audience Narrative	Bridge complex case timelines for families, search teams, and non-technical stakeholders.	Step-by-step 10yo accessible breakdown, Last Known Sighting (LKS) sequence, alternate route hazards.
2. Cellular & Telemetry Matrix	Reconstruct phone movement, tower handshakes, and egress dead zones.	Tower sector coverage radius, Round-Trip Time (RTT) distance arcs, highway dead-zone mapping.
3. Remote Sensing & CV Pipeline	Examine bare-earth terrain, vegetation anomalies, and apparel signatures.	LiDAR canopy stripping (sink depth >=15cm), HSV apparel color filtering, GPR / creek catchment targets.
4. Field & WebAR Deployables	Equip ground teams with real-time, interactive spatial visualization tools.	3D elevation cross-section charts, GeoJSON/KML placemark files, mobile WebAR 3D config packages.

Standardized Module Execution Protocols

Module 1: Multi-Audience Narrative Layer

Every dossier must open with a dual-narrative layout: (a) a formal operational timeline containing timestamps, weather, apparel, and witness credibility ratings; and (b) a clear, plain-language narrative suitable for young family members or public appeals. Alternate trajectories (wrong-way routes, shortcut hazards, and vehicle egress points) must be explicitly modeled alongside the zero-trace paradox.

Module 2: Cellular Telemetry & Telecommunication Matrix

Forensic phone analysis must evaluate both local tower sector pings and egress handoff patterns. Distance concentric arcs calculated via Round-Trip Time (RTT) are mapped against topographical obstacles. A key requirement is identifying terrain dead zones along mountain highways to determine if sudden ping termination signifies device destruction or vehicle movement into unserved valleys.

Module 3: Remote Sensing & Computer Vision Pipeline

Bare-earth LiDAR processing must strip vegetation layers to reveal ground micro-depressions (>=15 cm) such as tree wells, collapsed drainage culverts, and creek cut-banks. Computer vision scripts apply HSV color space filters (e.g. searching for jacket/cap apparel tones) and dimensional bounding boxes to prioritize high-confidence terrain anomalies.

Module 4: Interactive Field Assets & WebAR Deployment

To ensure analytical findings translate directly to field operations, each dossier must automatically compile and export: 2D elevation cross-section charts showing elevation drops and slope gradients, exportable `.kml`/`.geojson` vector overlays for ATAK/QGIS, and a mobile WebAR 3D scene configuration package for smartphone camera visualization.

OPERATIONAL MANDATE

Whenever a trajectory or movement dossier is requested for any subject, search managers must execute all four modules concurrently. This guarantees that searchers, data analysts, and family liaisons operate from a unified, multi-layered source of truth.